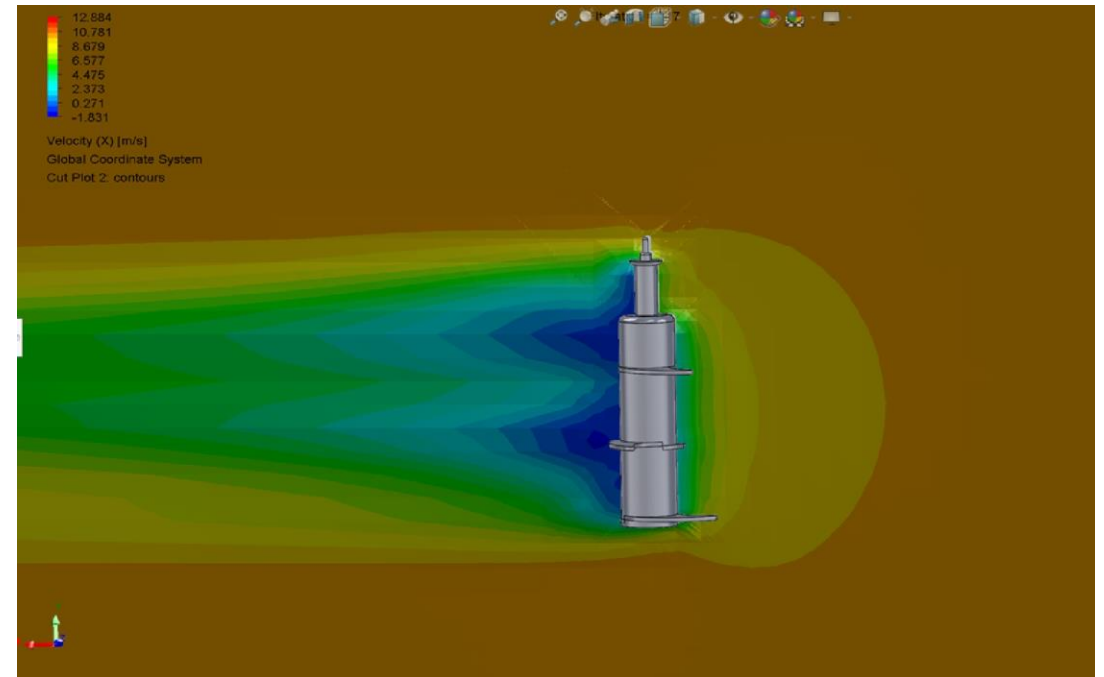
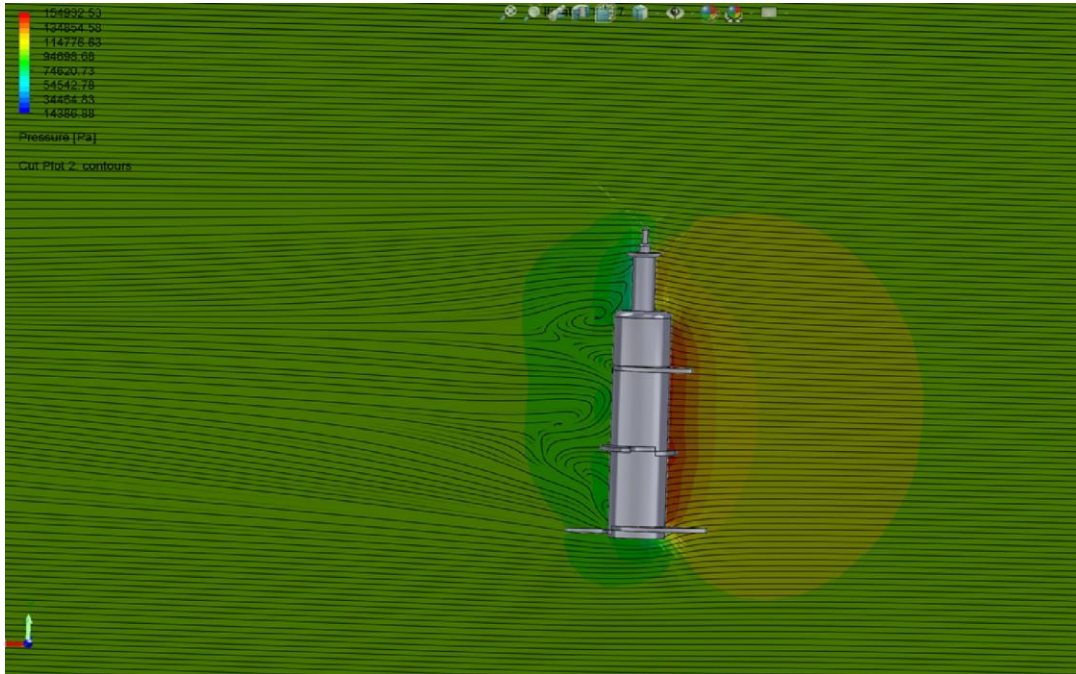


Blender Presentation

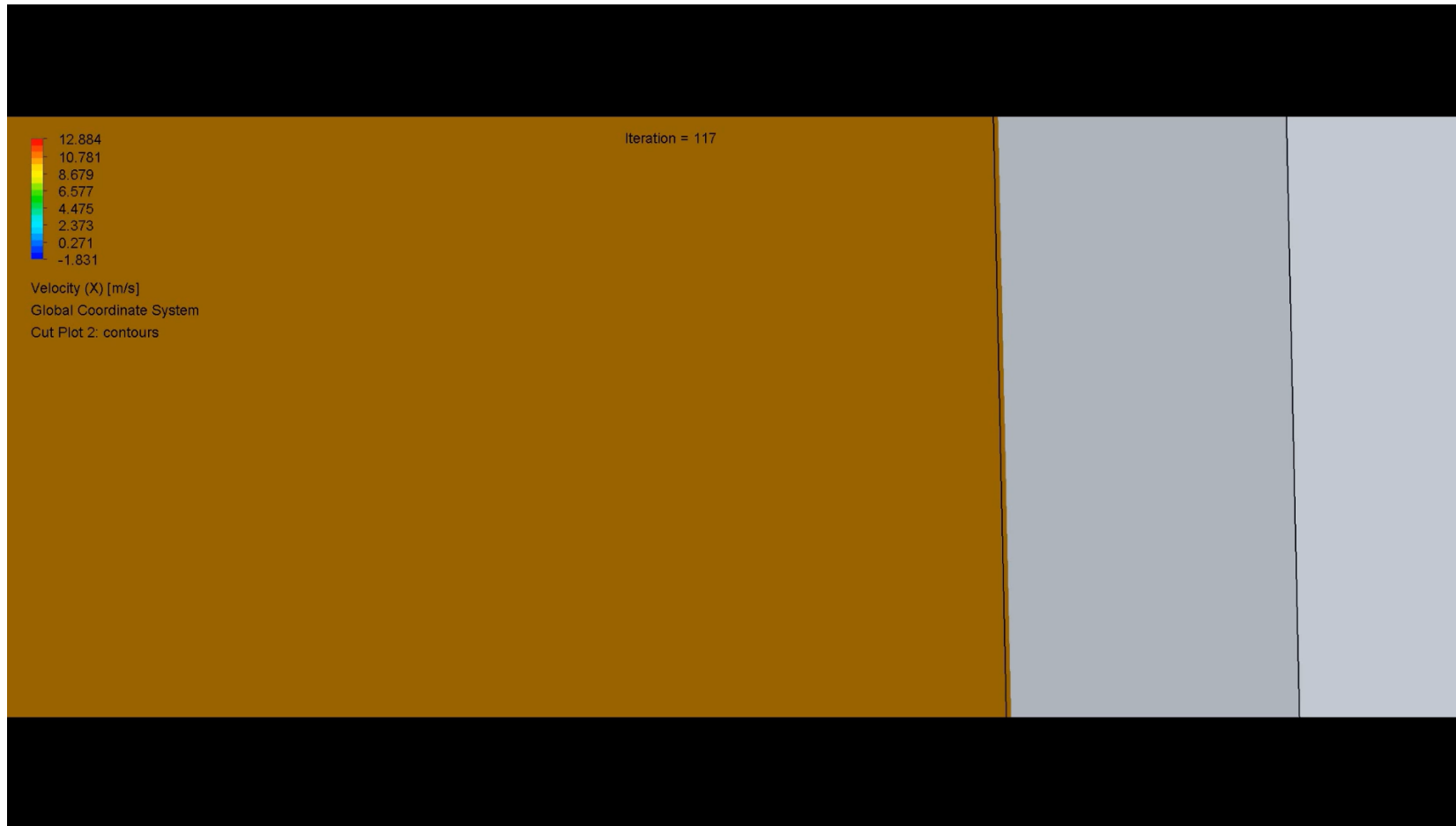
Carter Smith, Eliza Tabachnik, Akash Raghunathan,
Brenden Kline, Jake Haley, Opemipo Clement, Miles
Bryson

Analyses

Fluid Data Images

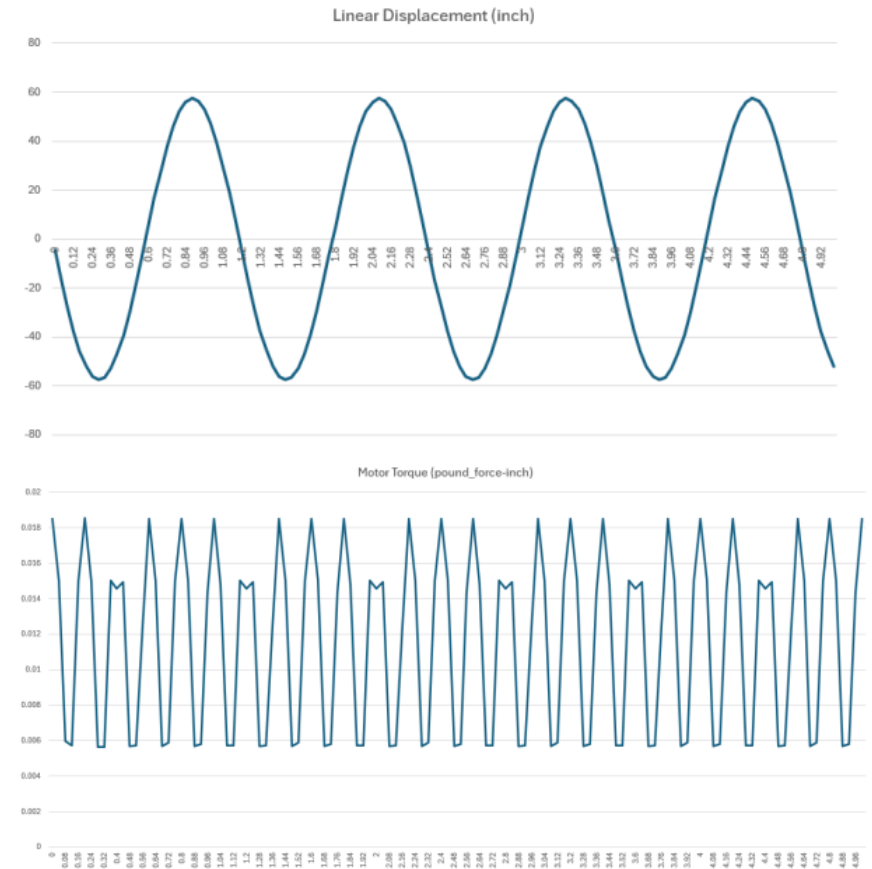


Water Velocity Animation

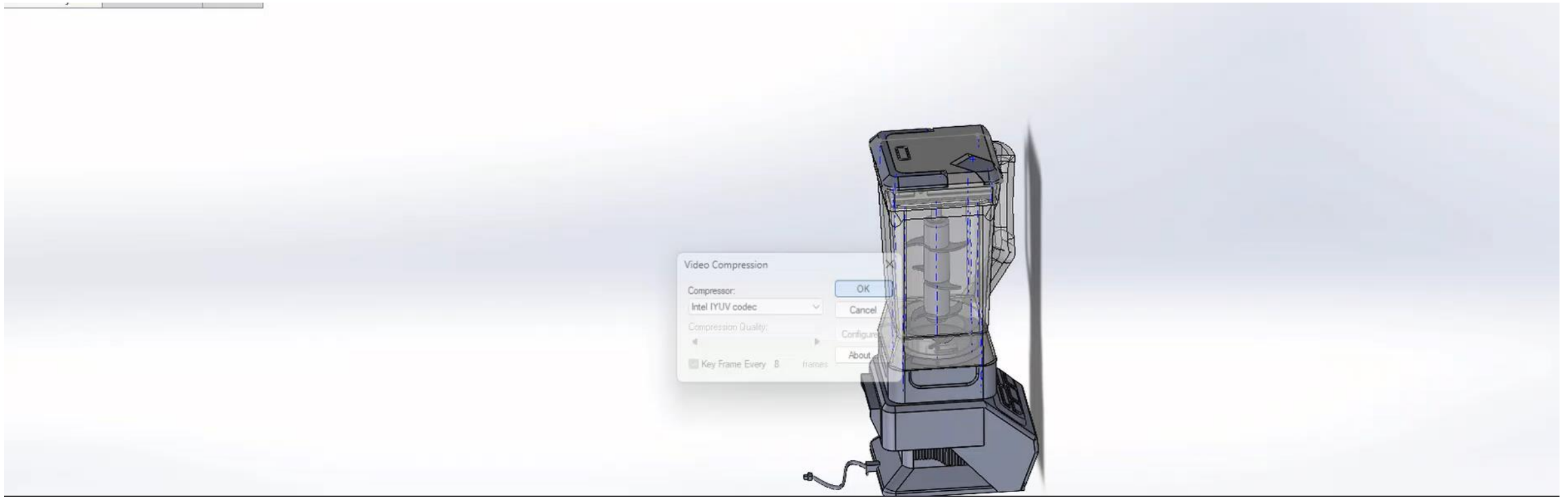


Motion Analysis

To analyze the motion of the blender, I created a motion study and added a motor to the driving gear of the blender. At first, I set the motor to rotate at a constant speed of 100 rpm and then used the Results and Plots function in SolidWorks to plot the linear displacement of the top face of one of the blades in the blade assembly relative to the driving gear. I then switched the motor features from rotation at a constant speed to oscillation at a displacement of 90 degrees, a frequency of 10 Hz, and a phase shift of 0 degrees. I then used the Results and Plots function to obtain the motor torque graph seen below

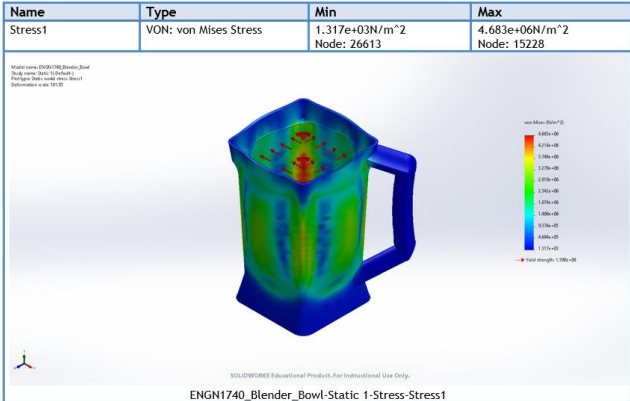


Motion Analysis Animation

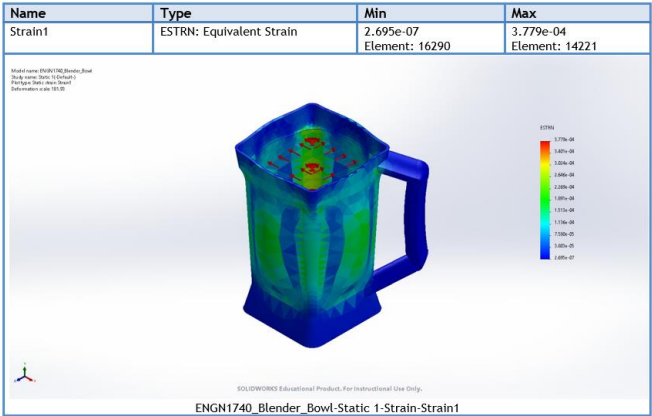


Structural Analysis

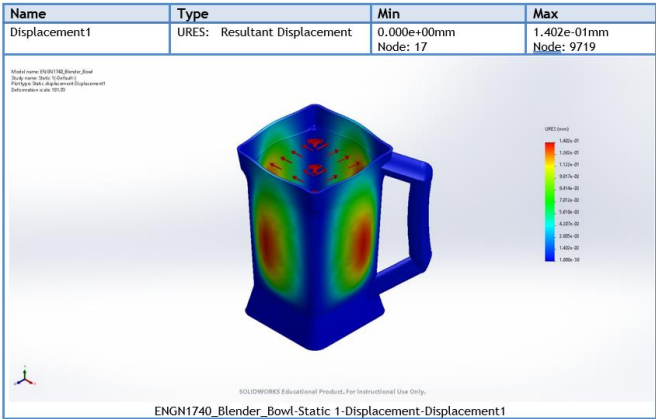
Study Results



Stress



Strain

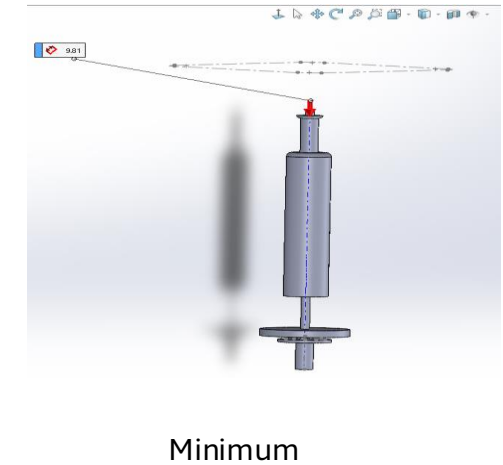
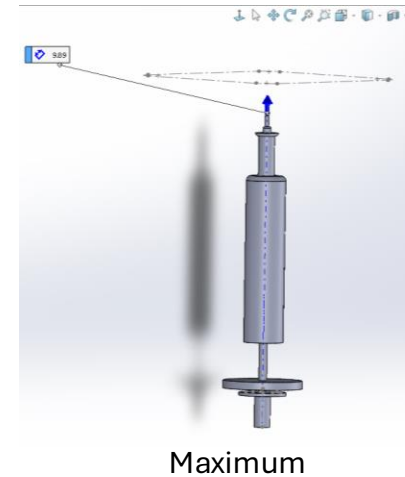


Displacement

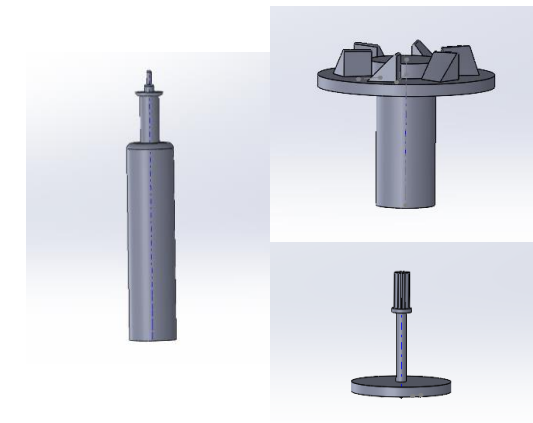
20,000 N/m² normal force applied to the walls and base, simulating normal pressure created by the spinning of the center blade. 6 N force applied to base, simulating the weight of a heavy smoothie. Nylon 6/10 material was used to simulate a worst-case scenario of material that deforms the easiest. The yield stress is not reached; therefore, the geometry of the blender bowl is structurally sound.

Tolerance Analysis

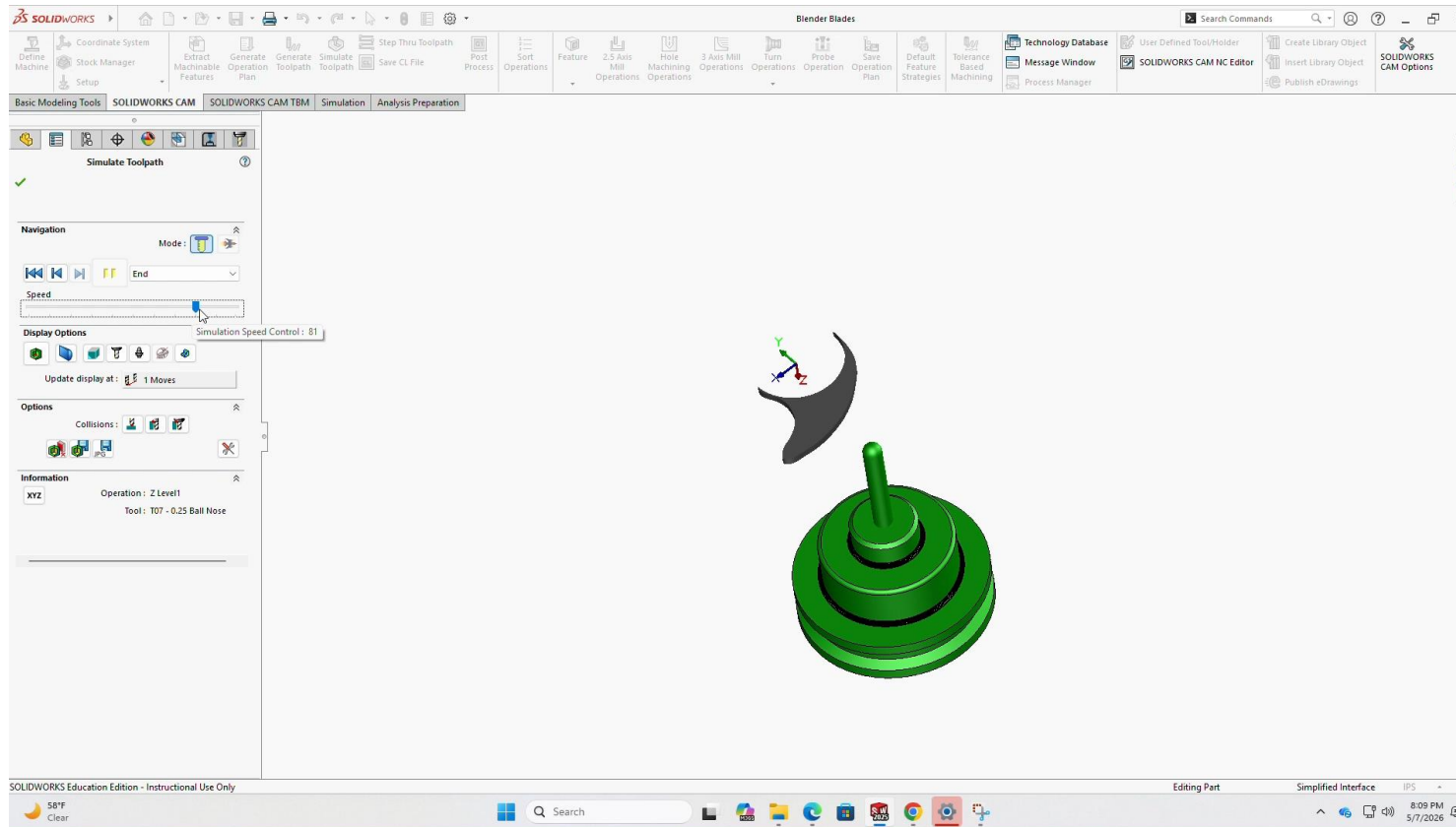
- Purpose: Analyzed the internal blade/drive assembly to see how part tolerances affect the final assembled measurement.
- Measurement: TolAnalyst reported a nominal value of 9.85 in, with a worst-case range of 9.81 to 9.89 in.
- Main contributors: The driving gear and starfish component were the largest contributors, each at 44.37%, while the blade stem contributed 11.27%.
- Conclusion: The stack-up stays close to nominal, so the assembly tolerance is acceptable for maintaining blade/drive alignment.



Individual
parts used



CAM Simulation for Blade

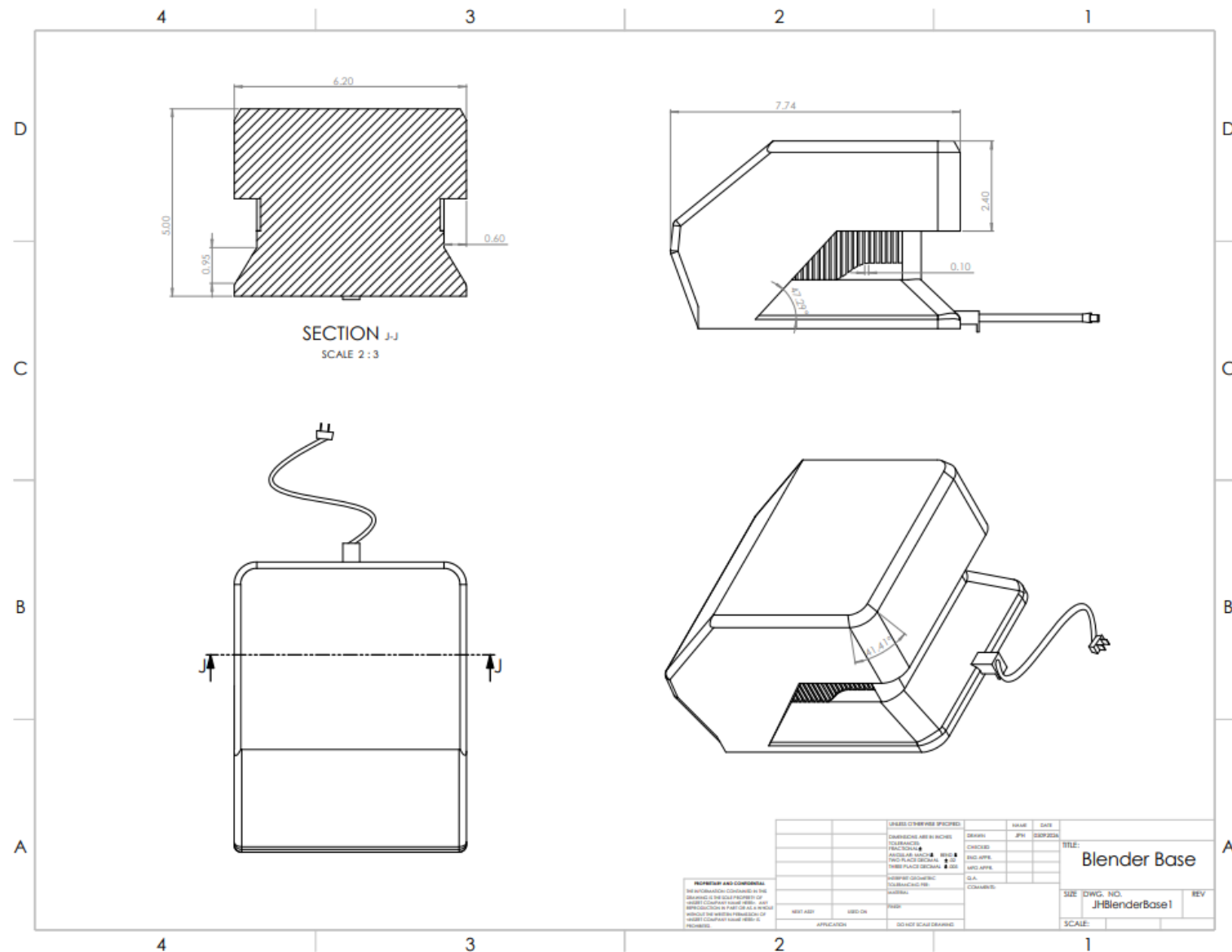


There is a two-setup milling operation on 6061-T6 aluminum using a mix of roughing/finishing methods, including Rough Mill, Contour, Area Clearance, and Z Level cycles. The toolpath utilizes primarily a 0.25" flat end mill (T01) for general clearing and a 0.25" ball nose (T07) for the finishing work. There are two machining orientations to ensure all features are reached.

Blender Drawings

[illegible]

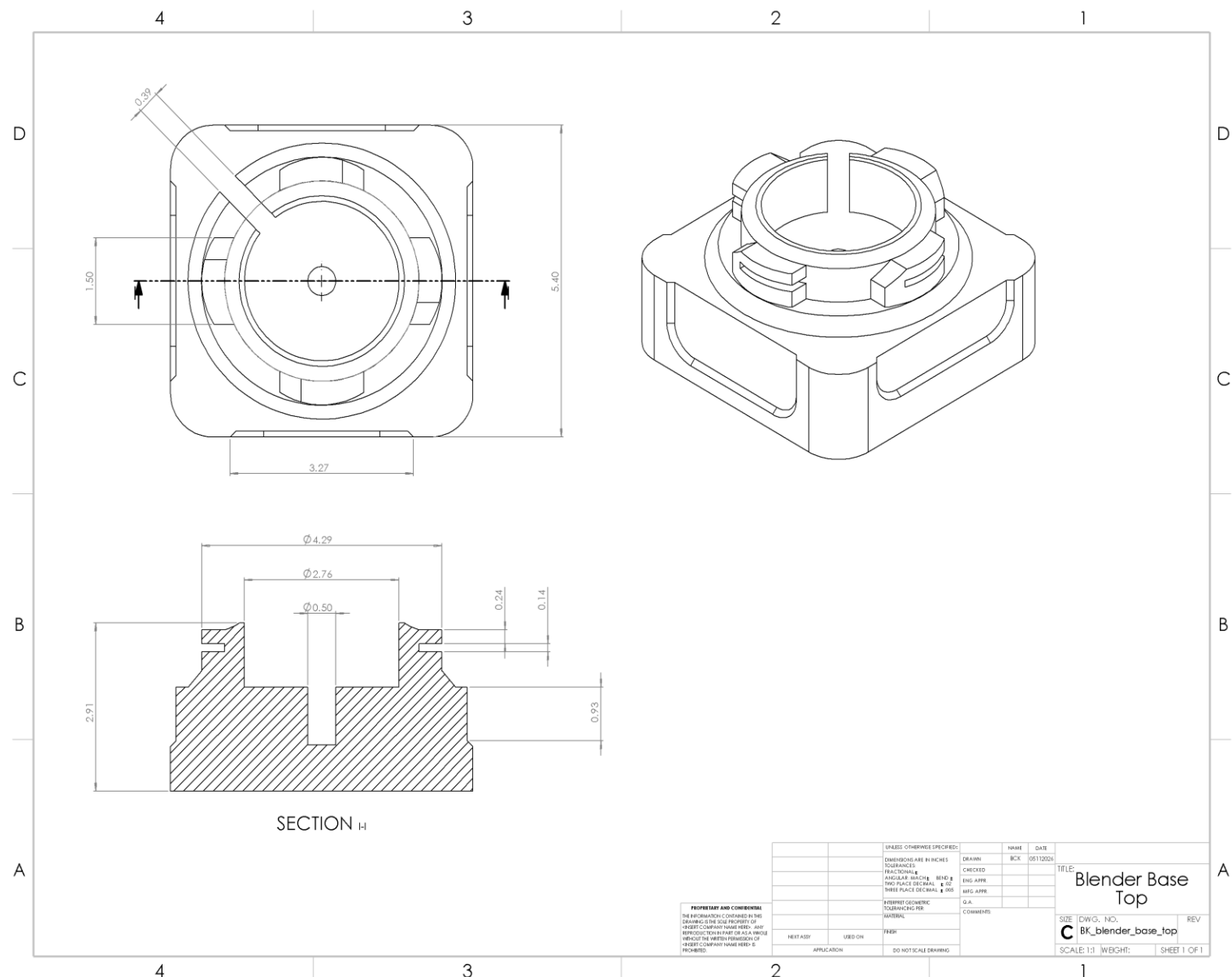
Blender Base Drawing



[illegible]

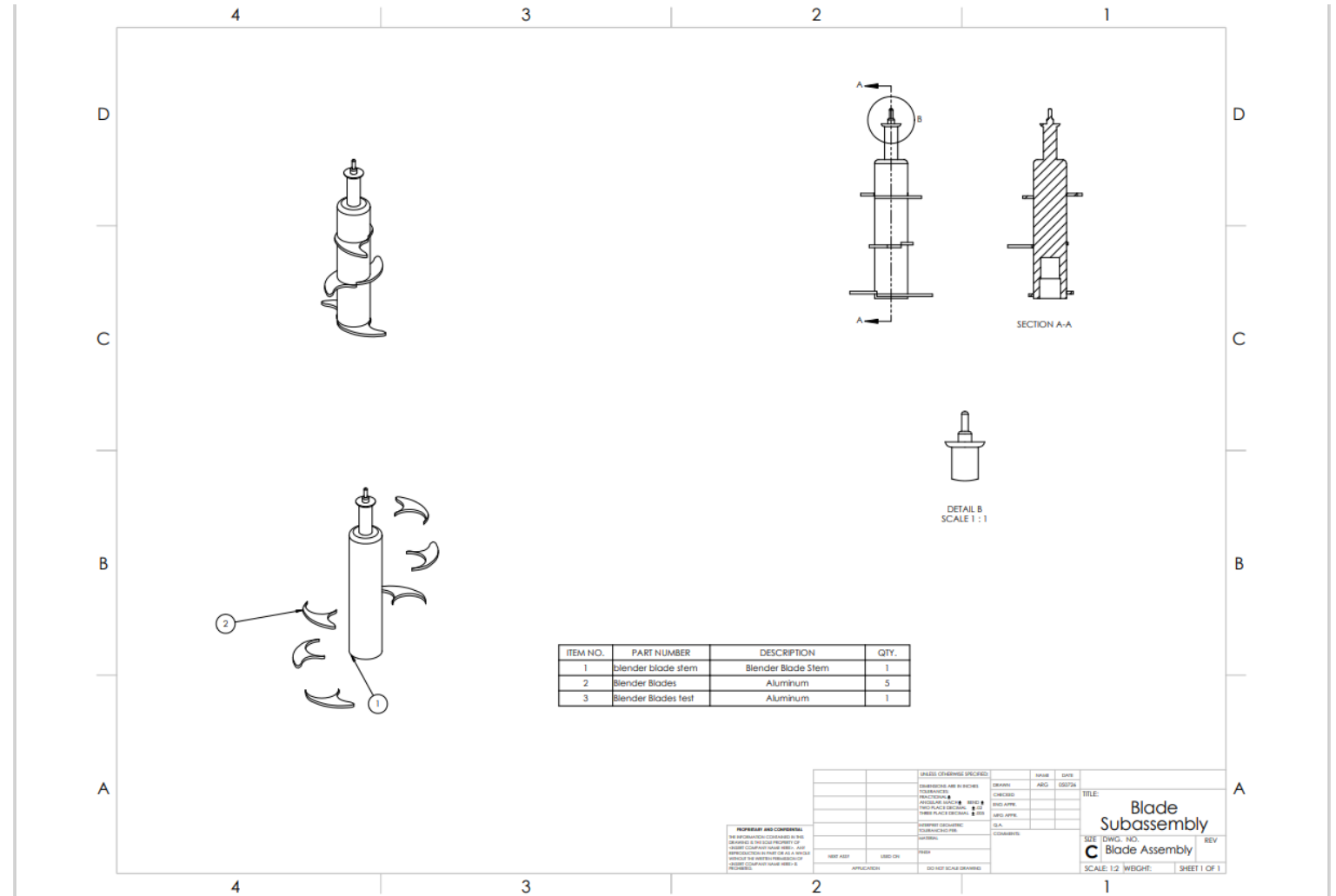
The technical drawing illustrates the design of a Blender Fan. It includes three main views: a front view, a side view, and an isometric view. The front view shows a rectangular unit with a width of 4.00 and a height of 1.70. It features a central section with horizontal slats, a top flange of 0.20, and a bottom flange of 0.09. A section line A-A is indicated. The side view shows a depth of 0.50 and a top flange of 0.20. A section line A-A is also indicated. The isometric view provides a three-dimensional perspective of the fan. A detailed section A-A is shown at the bottom right, with a scale of 3:1. This section reveals the internal structure, including a central core with a diameter of 0.05 and a top flange of 0.20. The section is labeled 'SECTION A-A' and 'SCALE 3:1'.

Blender Base Top Drawing

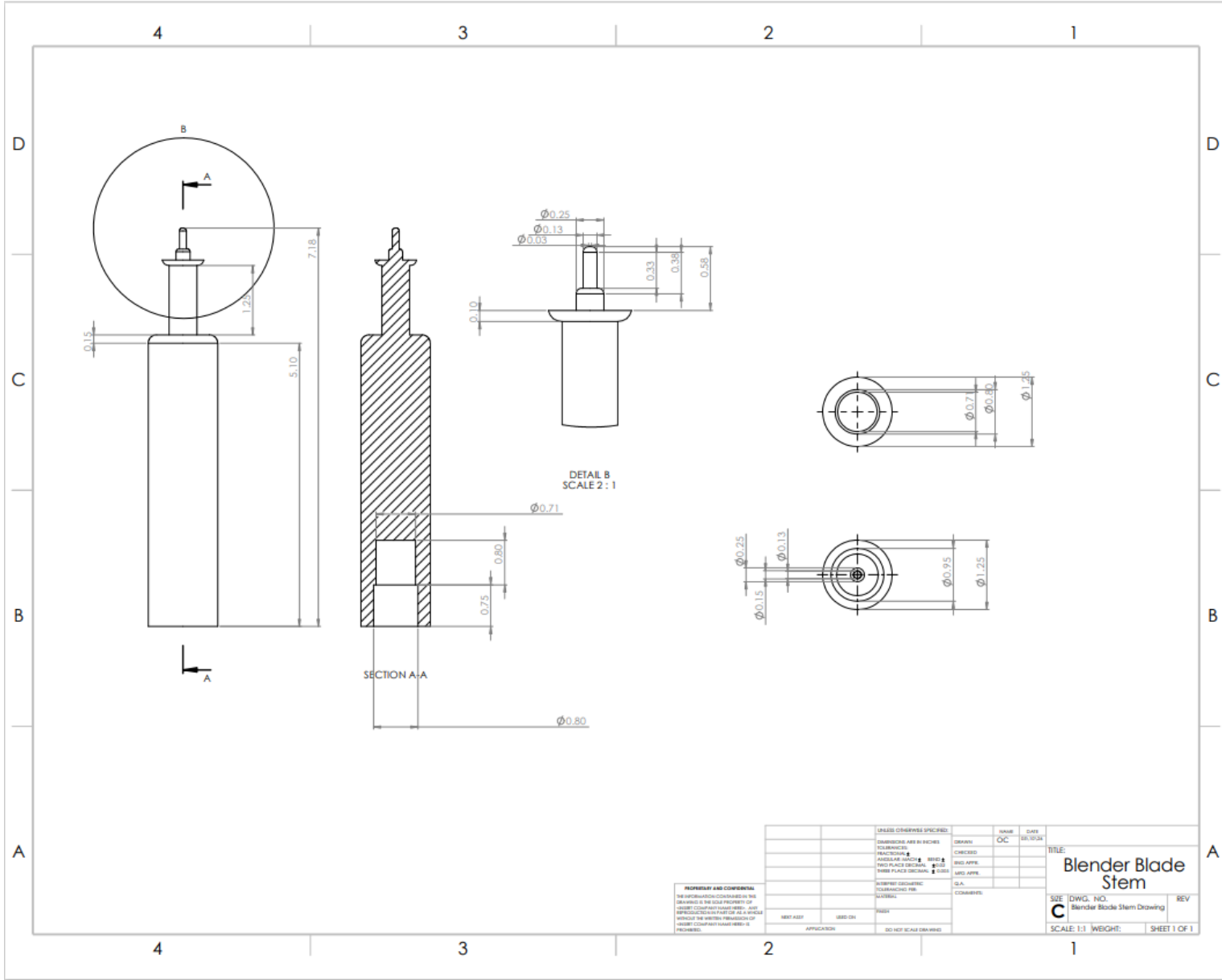


[illegible]

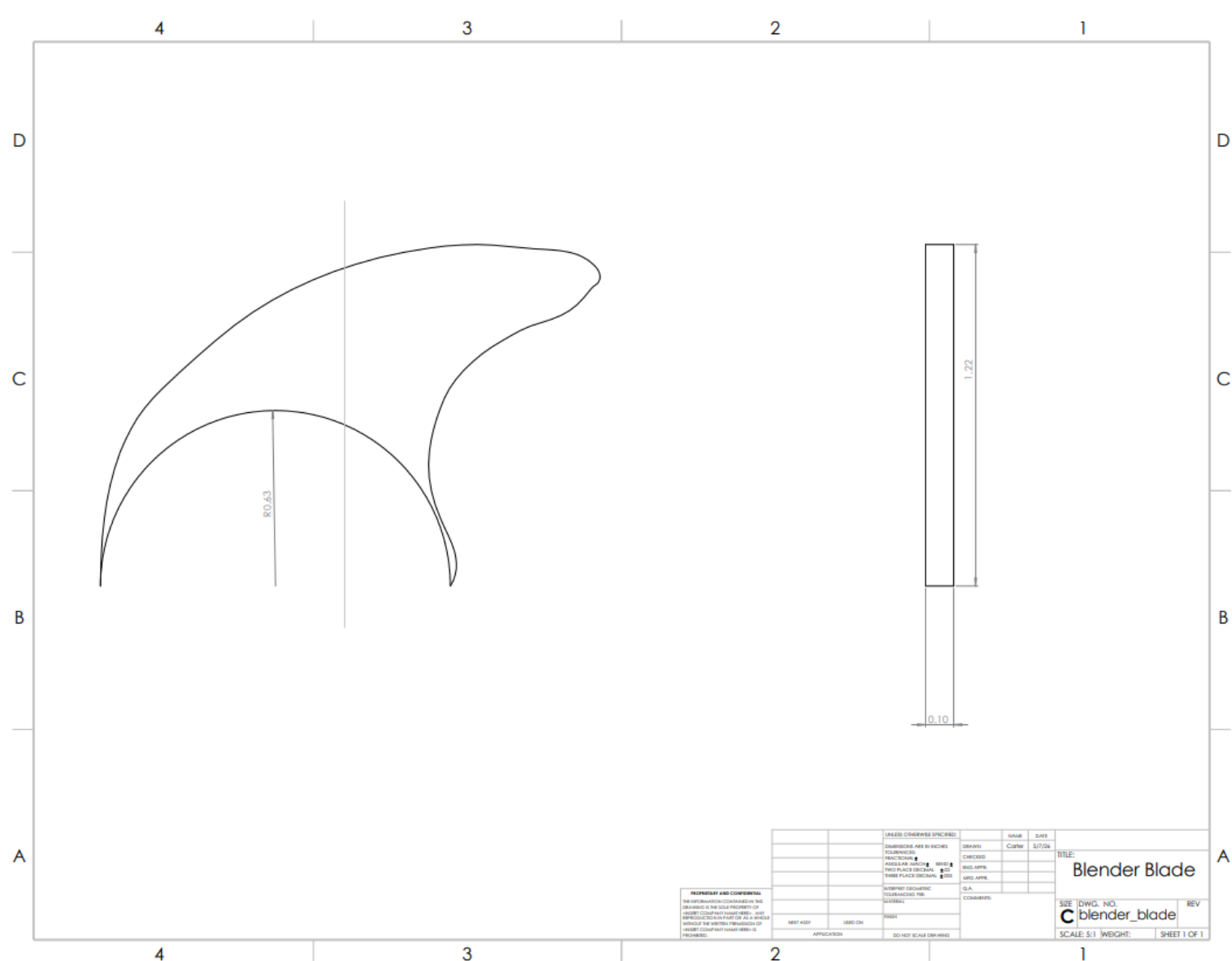
Blade Assembly Drawing



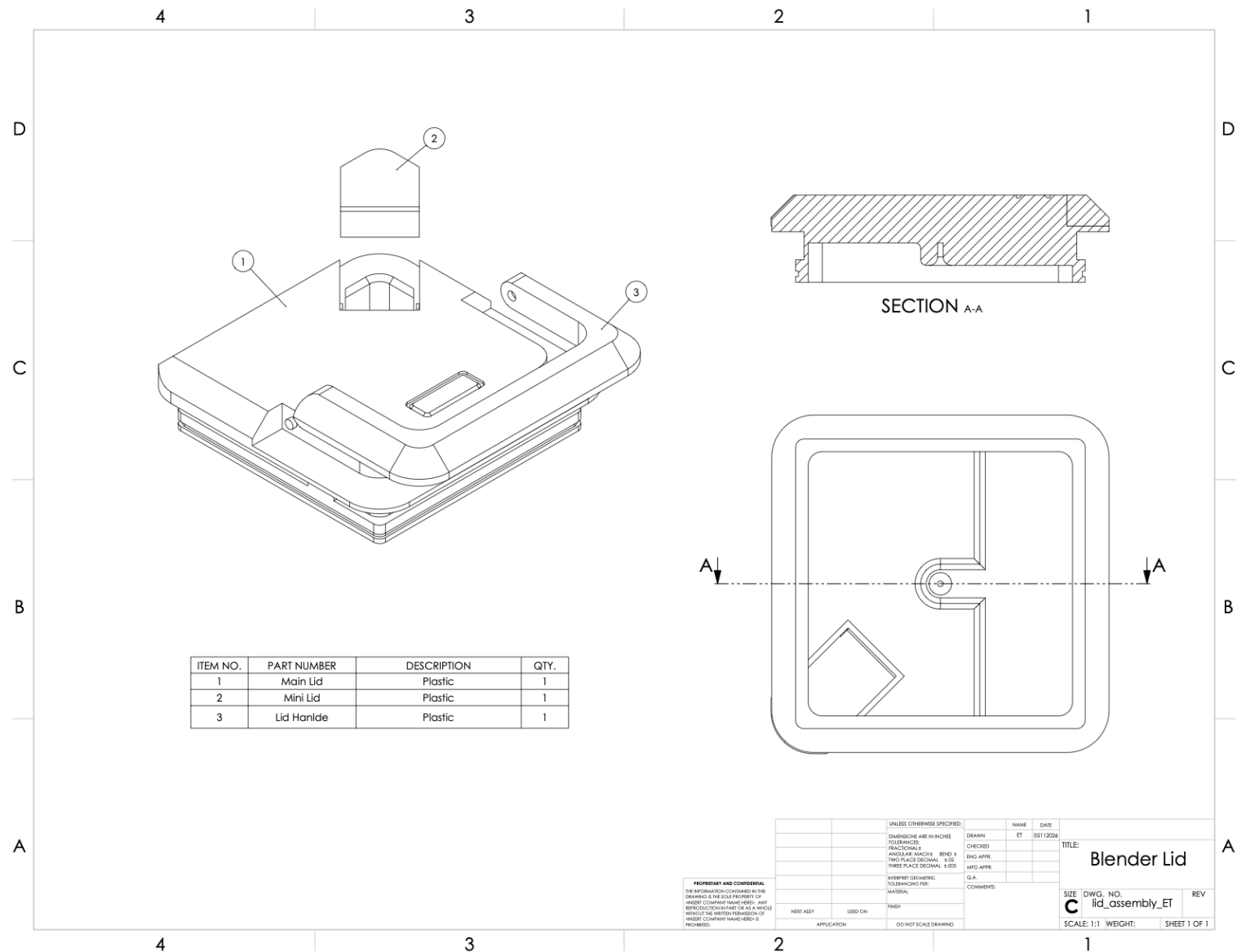
Blade Stem Drawing



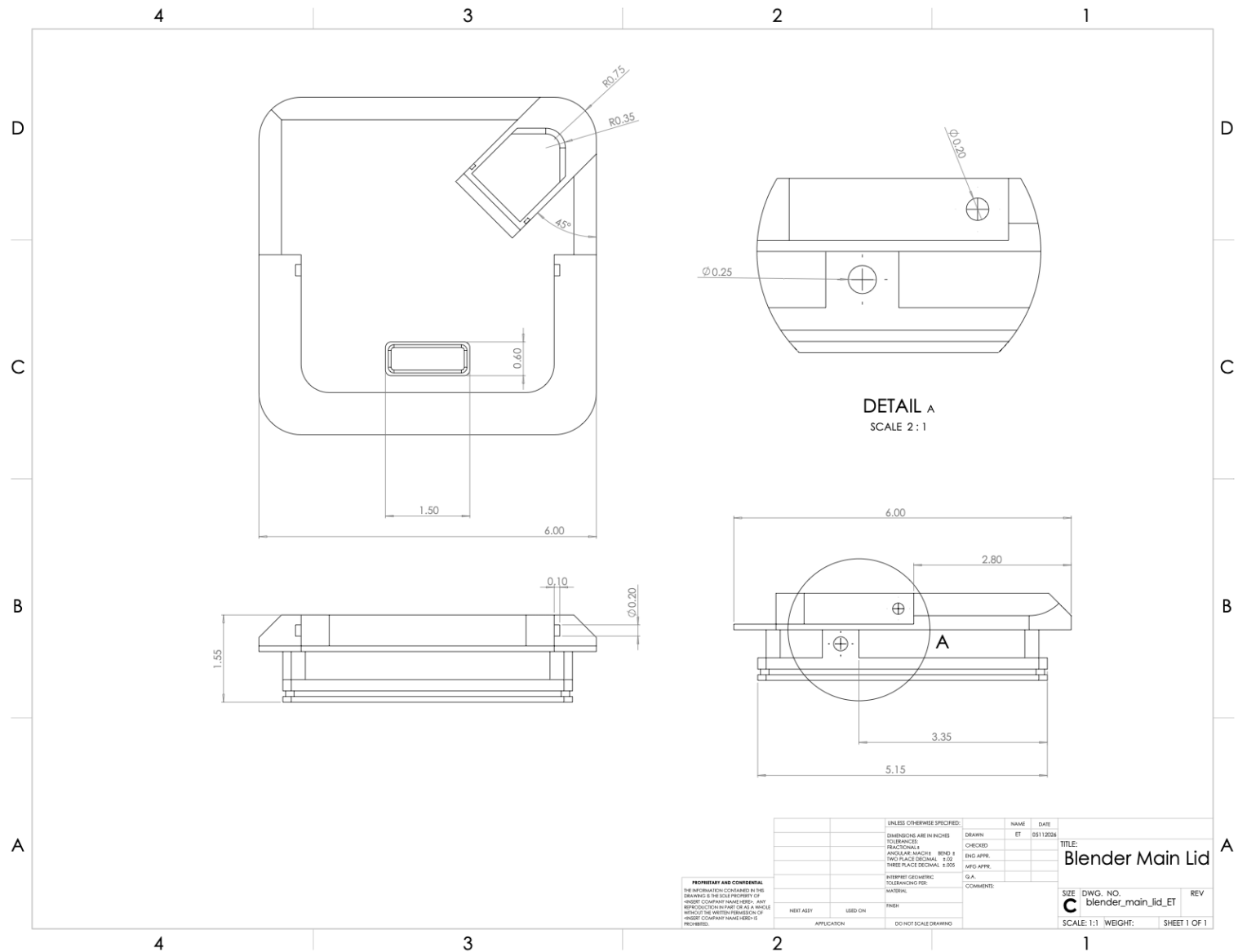
Blender Blade Drawing



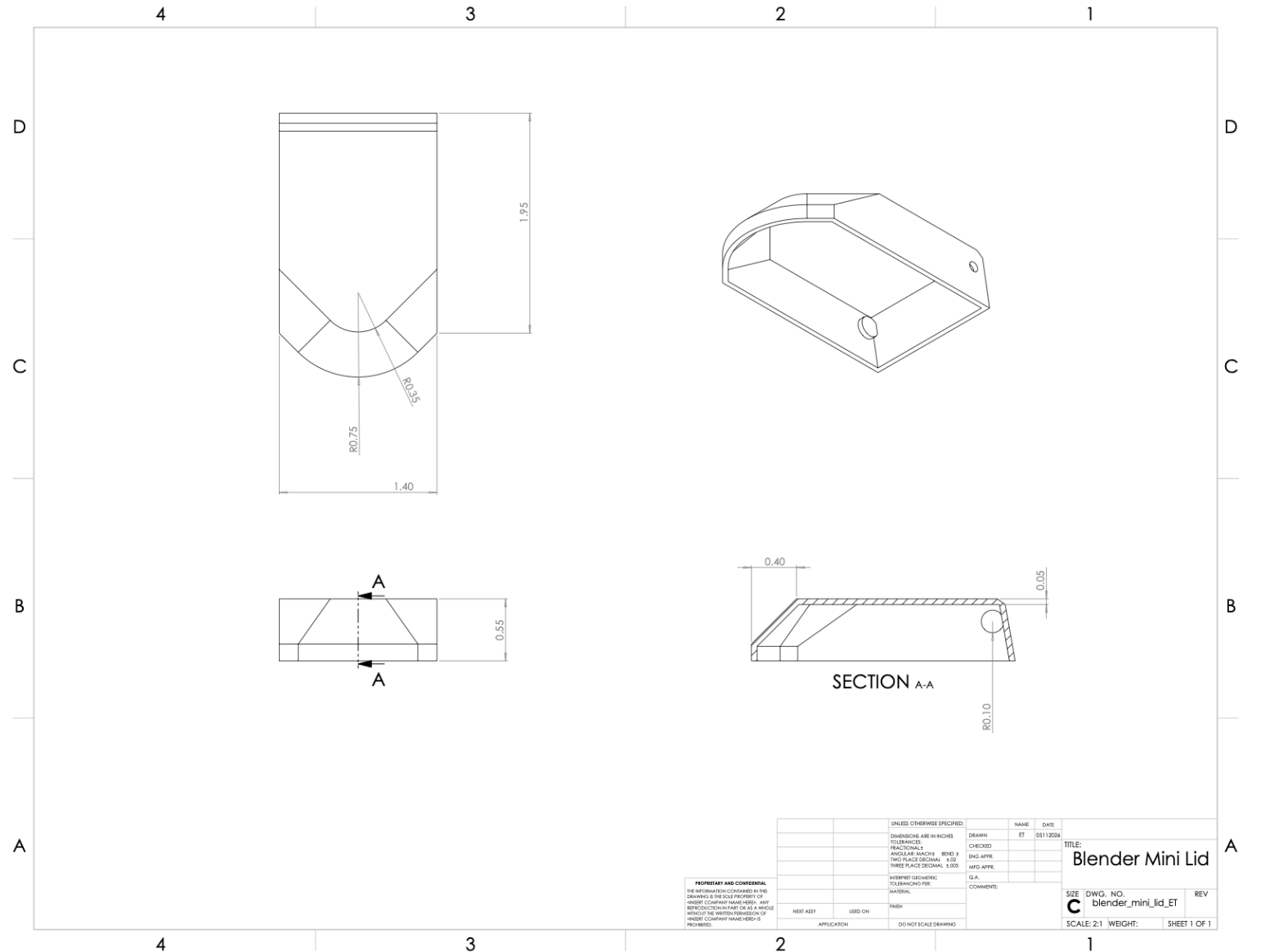
Lid Assembly Drawing



Blender Main Lid

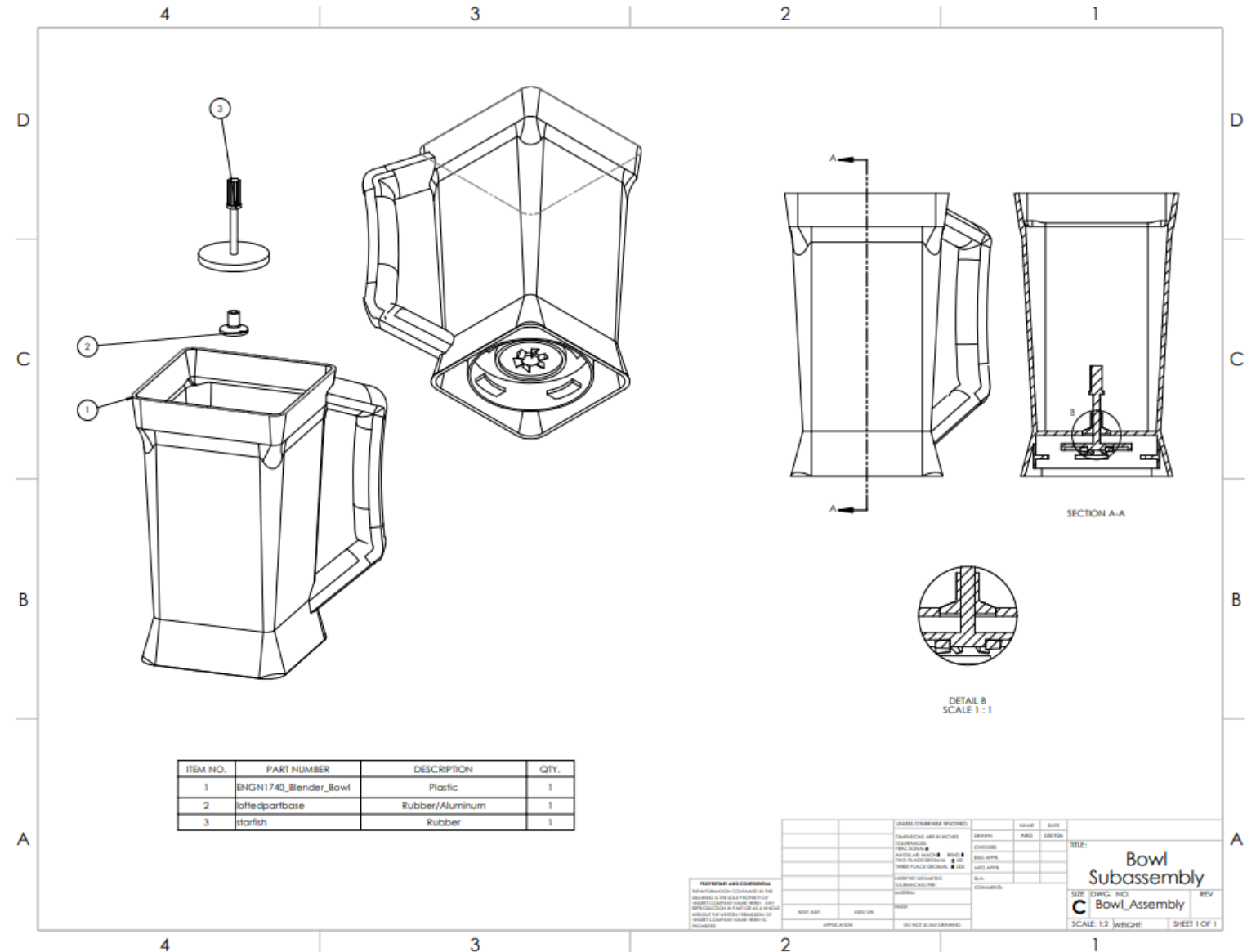


Blender Mini Lid

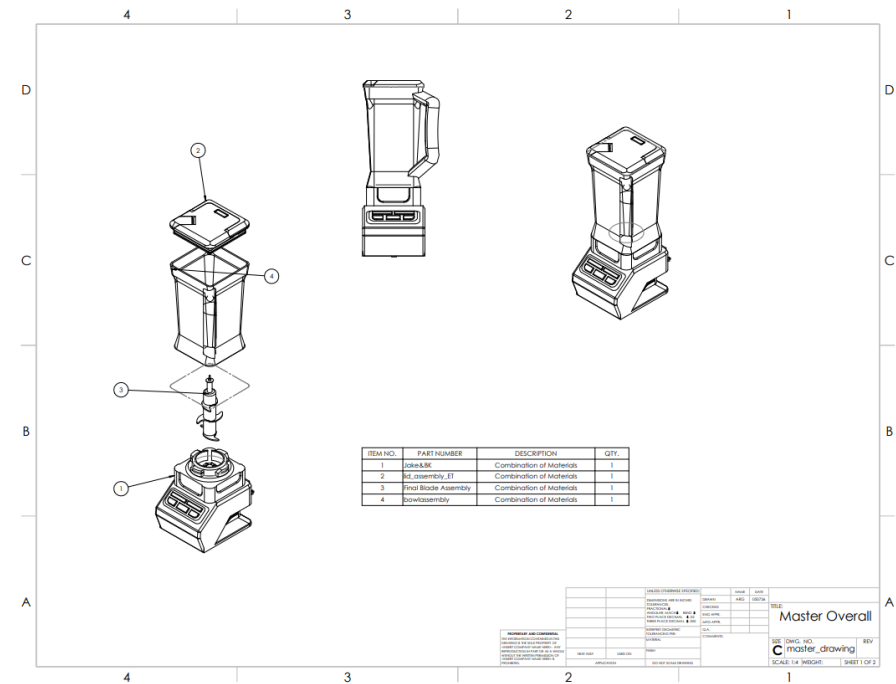
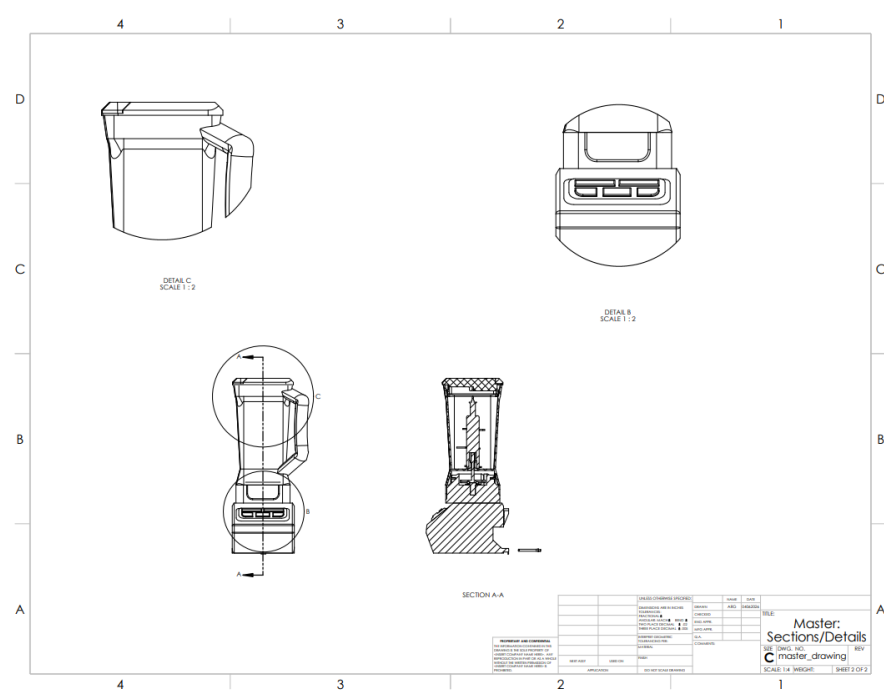


Technical drawing of a Blender Lid Handle. The drawing includes a top view showing a rectangular handle with rounded corners, a side view showing the handle's profile, and a cross-section labeled 'SECTION A-A' showing the handle's internal structure. Dimensions are provided: overall width 6.00, overall height 3.20, inner width 4.50, and a small detail of 0.10. A section line A-A is indicated. The drawing is framed by a coordinate system with letters A, B, C, D and numbers 1, 2, 3, 4.

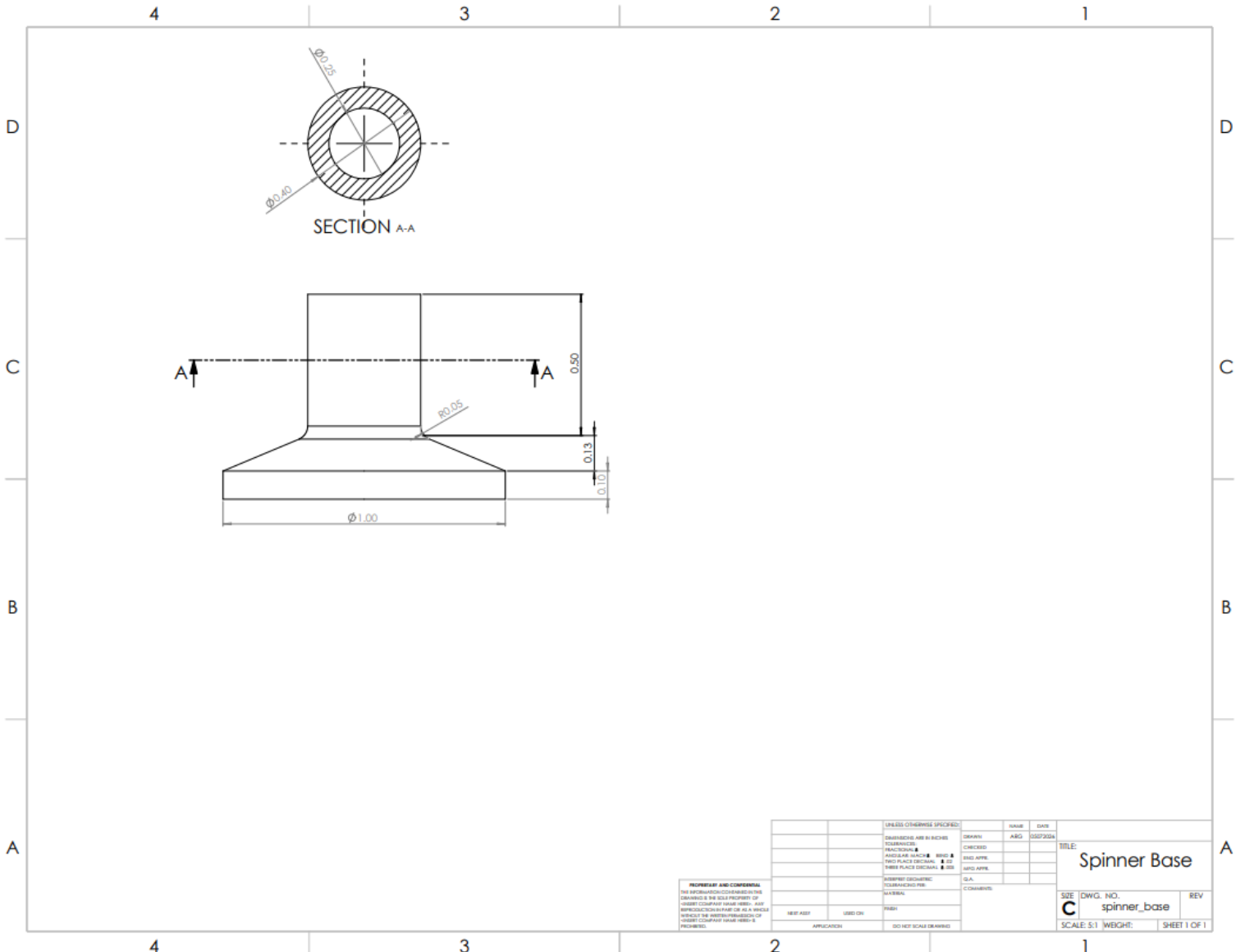
Bowl Assembly



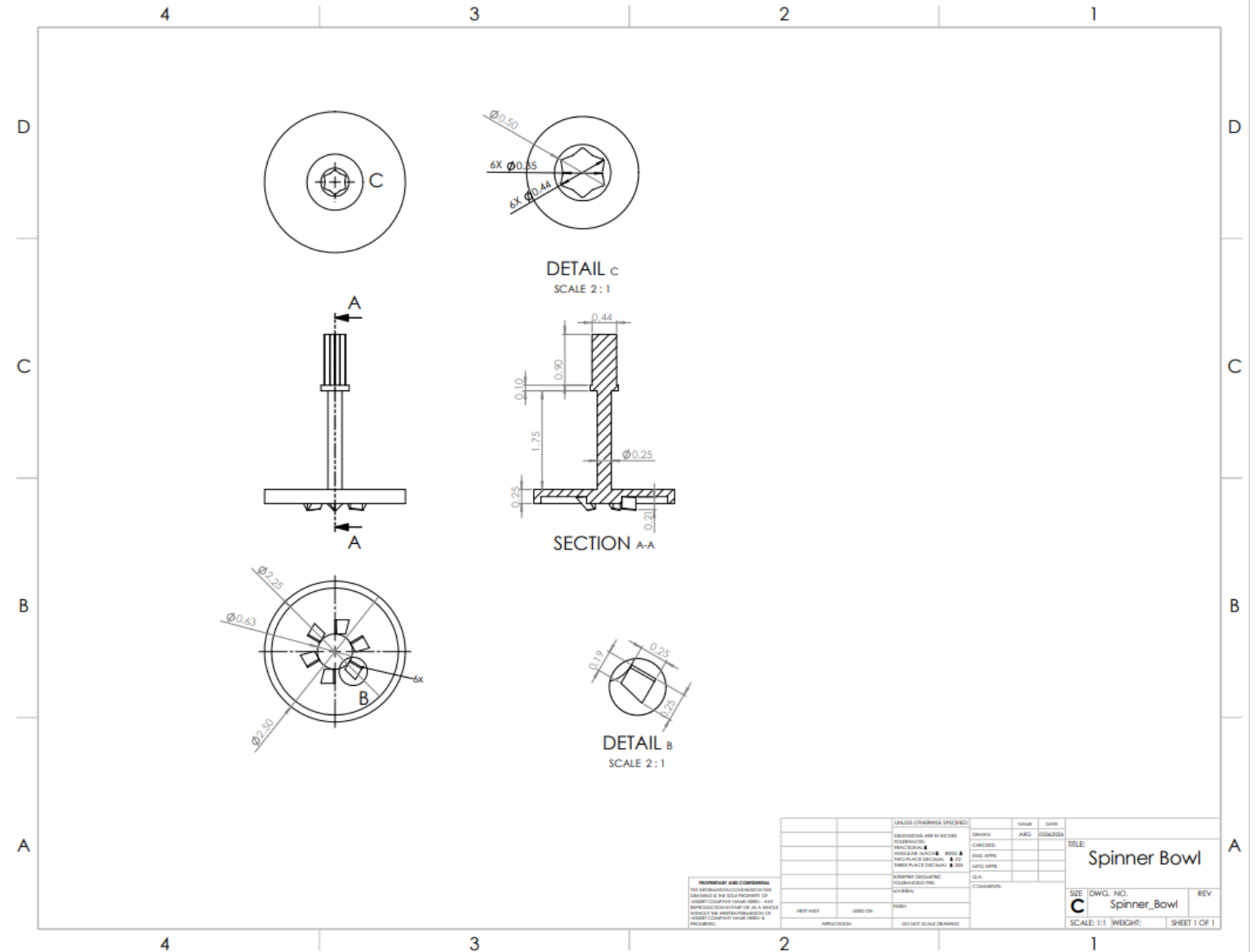
Master Assembly Drawings



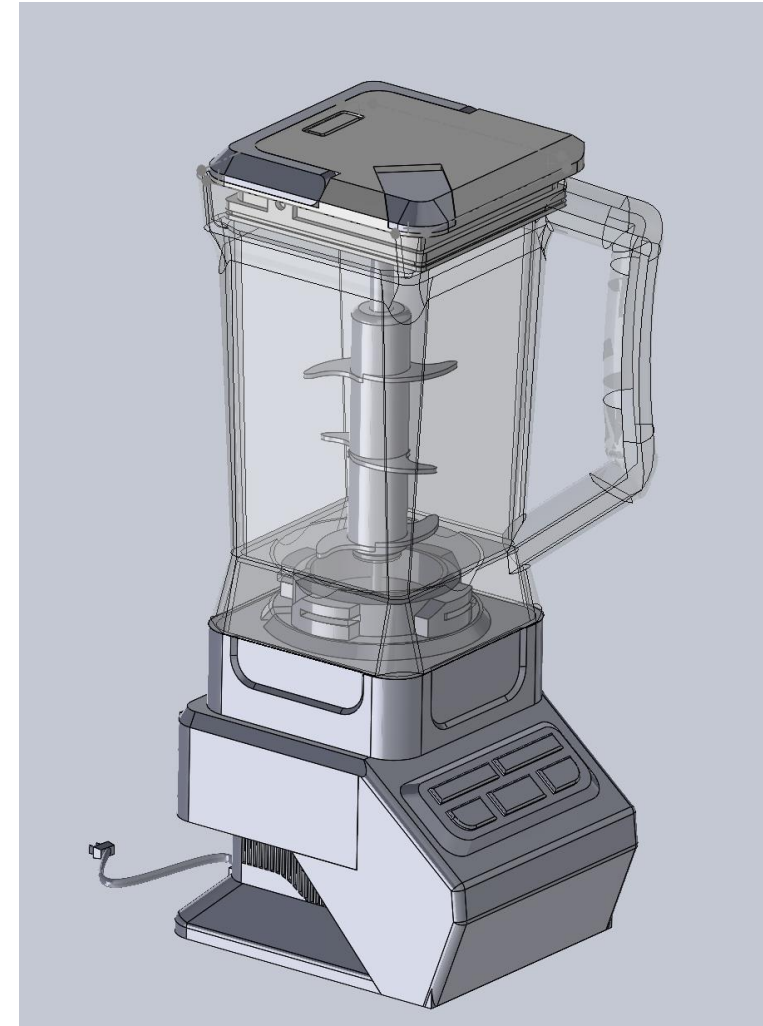
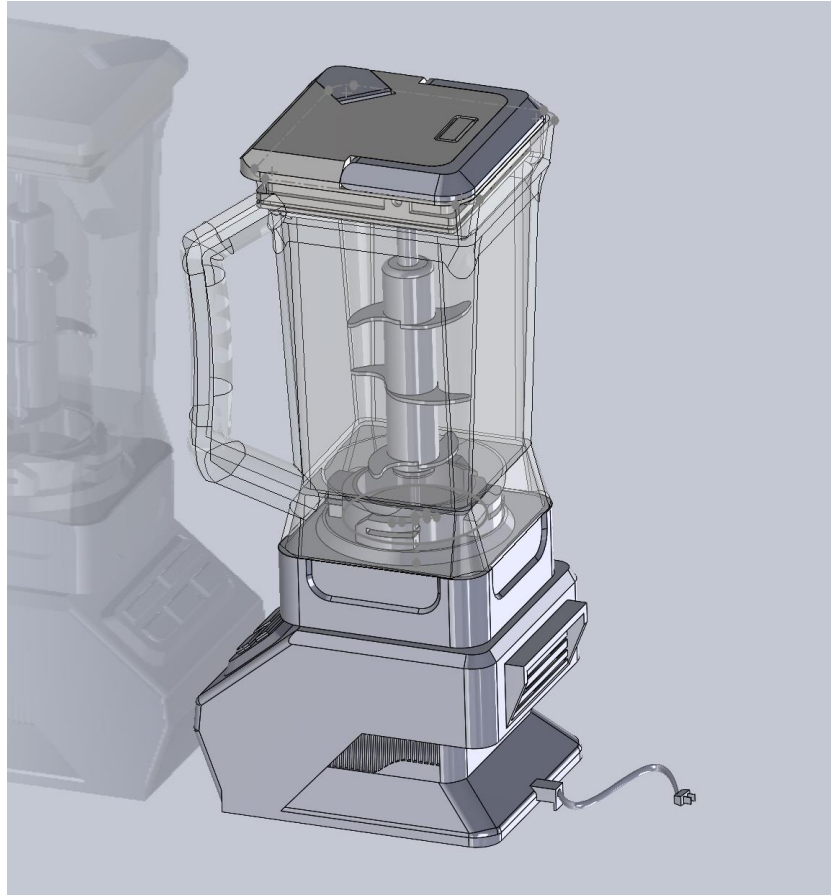
Spinner Base Drawing



Spinner Bowl Drawing



Main Assembly



Base Attachment

